Atomic Functions Beyond the Critical Dyadic Case

English reconstruction of Rvachev's chapter, repository synthesis, and new fractal–probabilistic frontiers

Research report prepared in the notation and house style of the `ProveIt` Fabius corpus

28 August 2026

Abstract

This report reconstructs in English, corrects at the level permitted by the surviving page images, and substantially expands the attached Russian material on Rvachev's atomic functions. The source begins with compactly supported solutions of differential–functional equations, reduces their existence to cancellation of zeros in an infinite Fourier product, specializes to the one-parameter densities

$$\widehat{h}_a(t) = \prod_{j\geq 1} \operatorname{sinc}(ta^{-j}), \qquad a > 1,$$

and treats the critical member $h_2 = \mathrm{up}$, polynomial reproduction by dyadic translates, and the first auxiliary Fup function. The exposition below preserves that organization and notation while separating four proof statuses: translated source results, established repository/literature connections, new theorems proved here, and explicit conjectures.

The principal new contribution is a complete geometric description of the separated regime $a > 2$. The set on which h_a fails to be locally polynomial is exactly the two-map Cantor attractor

$$K_a = \left\{ \sum_{j\geq 1} \varepsilon_j a^{-j} : \varepsilon_j \in \{-1, 1\} \right\}.$$

Every complementary gap of generation n carries a polynomial of exact degree n. This gives exact L^1 and L^∞ norms of all derivatives, the Hausdorff dimension $\log 2/\log a$, a closed geometric zeta function and its complete lattice of complex dimensions, an exact tube formula with nonconstant logarithmic oscillation, and a geometric law for the local polynomial degree. As $a \downarrow 2$, the rescaled degree converges to an exponential random variable, making the dyadic nowhere-analytic function a precise critical limit of full-measure polynomial splines.

A second new layer gives Bernoulli-number cumulants and complete Bell-polynomial moments for every $a > 1$, a Gaussian limit as $a \downarrow 1$, a uniform-law limit as $a \to \infty$, and an exact general-base negative-Laplace decomposition. The latter contains a genuinely one-periodic correction whose nonzero Fourier coefficients are

$$-\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{\log a}, \qquad \chi_k = \frac{2\pi \mathrm{i} k}{\log a},$$

thereby extending the dyadic Gamma–zeta correction to arbitrary base. A lower-W_{-1} saddle coordinate and an all-orders periodically modulated endpoint program are formulated. Reproducible Python code, numerical audits, figures, and data tables accompany the report.

Proof-status legend. [S] denotes a statement translated or cleanly reconstructed from the attached source; [R] denotes a connection already established in the surrounding repository or classical literature; [N] denotes a result proved in this report; and [C] denotes a plausible but presently unproved conjecture or research program. A statement may carry more than one marker when a source observation is sharpened here.

Contents

1 Scope, sources, and normalization

1.1 What was translated and how

The attached `rvachev.tex` is an OCR transcription of a Russian chapter titled *Atomic Functions*. It contains five sections: the general definition and Fourier existence criterion, the first-order two-shift equation, the special function up, polynomial representation by translates of up, and the beginning of a section on functions Fup_n. The attached PDF contains the surviving printed pages corresponding mainly to the latter part of the first-order analysis and the up section. The OCR is often reliable in prose and badly damaged in superscripts, factorials, binomial coefficients, and binary subscripts. Accordingly, this report follows three rules.

1. When the printed page makes a formula unambiguous, the formula is silently typeset in standard notation and the source equation number is retained in the discussion.
2. When the OCR and page image are structurally clear but individual indices are damaged, the invariant statement is reconstructed from the neighboring equations and verified independently.
3. When the surviving material does not determine every symbol, no fictitious exact transcription is supplied. Instead, a clean equivalent formulation is given and the uncertainty is recorded in section B.

The repository was audited recursively using its current document topology and its own provenance maps. The live corpus consists of the mathematical glossary, the primary exposition `Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex`, the non-elementarity paper, the Lean walkthrough, the canonical frontier volume, the thematically organized frontier drafts, and the vendored paper sources. The frozen `archive/` was cross-checked through its document-by-document manifest, which records the original location, first committed version, and whether a living copy supersedes it. The draft `MANIFEST.md` was used as the global index for the reorganized 28 August 2026 frontier tree, including the Fourier-decay corpus and the consolidated Thue–Morse, q-series, transform, inverse, sampling, representation, spectral-arithmetic, and multi-topic volumes. Current canonical TeX sources were read directly; superseded historical repetitions were used for provenance and nonduplication checks rather than counted as independent mathematical evidence.

The primary document already contains the formalized dyadic probability, Fourier, arithmetic, q-binomial, inverse, and lower-Lambert theories. The present report therefore concentrates on the translated *general-base* family h_a and on deductions that were not found as exact statements in that corpus. In particular, the report does not repackage the existing dyadic formula atlases, Fourier-decay audits, transform compilations, or lower-Lambert coefficient machinery. This is a nonduplication audit, not a claim of priority over the entire literature.

1.2 Fourier convention

Throughout,

$$\widehat{f}(t) = \int_{\mathbb{R}} \mathrm{e}^{\mathrm{i}tx} f(x)\,\mathrm{d}x, \qquad f(x) = \frac{1}{2\pi}\int_{\mathbb{R}} \mathrm{e}^{-\mathrm{i}tx}\widehat{f}(t)\,\mathrm{d}t, \tag{1}$$

and

$$\operatorname{sinc} z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

All logarithms are natural. Binary digits are indexed from the least significant digit: $p_j(m) \in \{0,1\}$ denotes the coefficient of 2^{j-1} in m, and $\mathrm{wt}_2(m) = \sum_j p_j(m)$.

1.3 Three related dyadic functions

The repository carefully separates the bounded Fabius distribution function F, the even compactly supported bump up, and the signed global extension. Only the first two are needed here. Their bridge is

$$\mathrm{up}(x) = F(1-|x|), \qquad F(x) = \mathrm{up}(x-1) \quad (0 \le x \le 1). \tag{2}$$

Thus the left-endpoint behavior of up is exactly the small-argument behavior of F. The general family h_a below contains up at $a = 2$ but does not, for arbitrary a, inherit all special dyadic identities of F.

2 Atomic functions and the Fourier-product criterion

2.1 The motivating compromise: local and universal

[S] The source begins from a practical tension in approximation theory. Algebraic and trigonometric polynomials are universal in the sense that the same class approximates functions of arbitrarily high smoothness, but they are nonlocal. Classical splines are local, and their compact support yields sparse or banded numerical matrices, but a fixed spline degree is not universal: increasing smoothness requires increasing degree. Rvachev's proposal is to use compactly supported C^∞ functions obtained as infinite convolutions of interval indicators. Their Fourier transforms are products

$$\widehat{\varphi}(t) = \prod_{k=1}^{\infty} \mathrm{sinc}(\alpha_k t), \qquad \alpha_k > 0, \qquad \sum_{k\ge 1} \alpha_k < \infty. \tag{3}$$

Finite convolutions give ordinary splines. Under the source's regularity hypotheses—in particular for the geometric scales used below—the infinite convolution gives a compactly supported infinitely differentiable limit. Mere summability of an arbitrary sequence (α_k) controls the support but, by itself, should not be read as a complete C^∞ criterion. The geometrically decreasing choice $\alpha_k = 2^{-k}$ produces up.

2.2 Differential–functional equations

Definition 2.1 (Atomic function, source normalization)**.** An *atomic function* is a nonzero compactly supported solution of

$$P(D)y(x) = \lambda \sum_{k=1}^{M} c_k y(ax - b_k), \qquad a > 1, \qquad D = -\mathrm{i}\frac{\mathrm{d}}{\mathrm{d}x}, \tag{4}$$

where P is a polynomial and the right-hand side is a finite dilation–translation operator.

Put

$$\mathcal{E}(t) = \frac{1}{a} \sum_{k=1}^{M} c_k \mathrm{e}^{\mathrm{i}tb_k/a}. \tag{5}$$

Taking Fourier transforms in (4) gives the exact scalar recursion

$$P(t)Y(t) = \lambda \mathcal{E}(t) Y(t/a). \tag{S3.2}$$

Iteration yields

$$Y(t) \prod_{j=0}^{n} P(ta^{-j}) = \lambda^{n+1} \prod_{j=0}^{n} \mathcal{E}(ta^{-j}) Y(ta^{-n-1}). \tag{S3.7}$$

This is the basic reason Fourier analysis is as effective for atomic equations as it is for constant-coefficient ODEs.

2.3 Zero cancellation and normalized solutions

[S] Suppose first that $Y(0) = \int y = 1$. Then the zero of lowest order of P at the origin must match the zero of the same order of $\mathcal{E}$, and the normalization fixes λ. Away from the origin, every zero introduced by the denominator in the infinite iteration must be canceled by a scaled zero of the exponential polynomial $\mathcal{E}$. The source phrases this as an injection between the multisets of zeros of P and $\mathcal{E}$ along multiplicative a-orbits.

Theorem 2.2 (Fourier-product existence criterion, cleaned source statement)**.** *Within the source class and its stated growth hypotheses, let A be the multiset of zeros of P and B the multiset of zeros of $\mathcal{E}$. Assume the origin multiplicities and normalization are compatible. A normalized nonzero compactly supported solution of* (4) *exists if and only if the zeros of P can be assigned, with multiplicity, to distinct zeros of $\mathcal{E}$ of the form $a^{-m}\alpha$, $m \in \mathbb{N}_0$, along their dilation orbits. Under this cancellation condition,*

$$Y(t) = \prod_{j=0}^{\infty} \frac{\lambda \mathcal{E}(ta^{-j})}{P(ta^{-j})}, \tag{S3.8}$$

with removable singularities filled in, and y is recovered by Fourier inversion.

Proof architecture. Necessity follows from (S3.7): for fixed t, the terminal factor tends to $Y(0) = 1$, so an uncanceled denominator zero is impossible. For sufficiency, the cancellation makes (S3.8) entire. The factors approach 1 uniformly on compact sets, giving local uniform convergence. The exponential type is controlled by the summable translation scales, while repeated use of the functional equation gives decay faster than every power on the real axis. Paley–Wiener then gives compact support, and Fourier inversion verifies the equation. □

Warning 2.3. The OCR preserves the zero-orbit criterion and product, but not every side condition from the preceding pages with equal reliability. Accordingly, theorem 2.2 records the source mechanism inside its original analytic class; it is not an assertion that an arbitrary polynomial and exponential mask satisfying only the displayed zero assignment automatically produce a C^∞ compactly supported inverse transform.

[R] In modern language this theorem belongs to the same broad family as refinement and two-scale equations: existence is encoded by a mask, dilation, and cancellation of spectral obstructions. The Daubechies–Lagarias theory studies finite masks and local regularity by matrix products; Rvachev's equation differs in that a differential operator appears on the left, but the Fourier recursion has the same renormalization structure [10, 11].

2.4 The derivative ladder when the mass vanishes

[S] If $Y(0) = 0$ and the zero has order r, write $Y(t) = t^r G(t)$ with $G(0) = 1$. The normalized function corresponding to G solves the same cancellation problem with the eigenvalue rescaled by a^{-r}, while the original compactly supported solution is a constant multiple of the rth derivative of that normalized solution. Consequently the admissible eigenvalues form a geometric ladder and the derivative solutions are linearly independent. This observation foreshadows the exact derivative formulas for h_a and up.

3 The first-order family h_a

3.1 Reduction of the two-shift equation

[S] For $P(t) = -it + \alpha$ and two shifts, translation and scaling reduce the equation to

$$y'(x) + \alpha y(x) = \lambda\big(y(ax-1) + c\, y(ax+1)\big). \tag{S3.22}$$

When $\alpha \neq 0$, the zero-cancellation condition selects a countable family of coefficients c; the source gives them explicitly through logarithms of the two-term exponential mask. The central case for atomic-function theory is $\alpha = 0$, where normalization forces $c = -1$ and $\lambda = -a^2/2$.

Definition 3.1 (Generalized Rvachev density)**.** For $a > 1$, define

$$\Phi_a(t) = \prod_{j=1}^{\infty} \operatorname{sinc}(ta^{-j}), \qquad h_a(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \mathrm{e}^{-\mathrm{i}tx} \Phi_a(t)\, \mathrm{d}t. \tag{S3.24–S3.25}$$

The Fourier recursion is

$$\Phi_a(t) = \operatorname{sinc}(t/a)\Phi_a(t/a), \tag{S3.28}$$

and differentiation of the corresponding integral refinement gives

$$h_a'(x) = \frac{a^2}{2}\big(h_a(ax+1) - h_a(ax-1)\big) = -\frac{a^2}{2}\big(h_a(ax-1) - h_a(ax+1)\big). \tag{S3.23}$$

Equivalently,

$$h_a(x) = \frac{a}{2} \int_{ax-1}^{ax+1} h_a(u)\, \mathrm{d}u. \tag{6}$$

The last formula immediately gives $0 \leq h_a(x) \leq a/2$ once the probabilistic interpretation is known.

3.2 Infinite convolution and support

Theorem 3.2 (Probability representation, source)**.** *Let $U_1, U_2, \dots$ be independent random variables uniformly distributed on $[-1, 1]$ and put*

$$X_a = \sum_{j=1}^{\infty} a^{-j} U_j. \tag{S3.26}$$

Then Φ_a is the characteristic function of X_a, h_a is its probability density, and

$$\operatorname{supp} h_a = [-b, b], \qquad b = \sum_{j \geq 1} a^{-j} = \frac{1}{a-1}. \tag{S3.27}$$

The density is even, nonnegative, C^∞, and has total mass 1.

Proof. The characteristic function of $a^{-j}U_j$ is $\operatorname{sinc}(ta^{-j})$, so independence gives the product. Almost surely $|X_a| \leq b$, and every point of $[-b, b]$ is approximated by finite choices of the coordinates, so the topological support is the entire interval. Along the real axis the product decreases faster than every inverse power. Indeed, if $N \asymp \log_a |t|$, then the first N factors are bounded by $a^j/|t|$, giving

$$|\Phi_a(t)| \leq \exp\left(-\frac{(\log |t|)^2}{2 \log a} + \mathcal{O}(\log |t|)\right). \tag{7}$$

Thus every $t^m \Phi_a(t)$ is integrable and $h_a \in C^\infty$. Compact support also follows from the exponential type $\sum a^{-j} = b$ by Paley–Wiener. □

[R] This is an infinite convolution in the sense of Jessen and Wintner [1]. It is simultaneously a fixed point of a smoothing transform: X_a has the same law as $a^{-1}(U + X_a')$, where X_a' is an independent copy. The resulting integral refinement (6) is the continuous-mask analogue of a subdivision equation.

3.3 Finite sinc products as nonuniform box splines

For $N \geq 1$, set

$$\Phi_{a,N}(t) = \prod_{j=1}^{N} \operatorname{sinc}(ta^{-j}), \qquad X_{a,N} = \sum_{j=1}^{N} a^{-j} U_j, \tag{8}$$

and let $h_{a,N}$ be the density of $X_{a,N}$. It is a compactly supported piecewise polynomial of degree $N-1$, a one-dimensional box spline with nonuniform directions $a^{-1}, \dots, a^{-N}$. If $w_j = a^{-j}$, $W_N = \sum_{j \leq N} w_j$, and $W_S = \sum_{j \in S} w_j$, then inclusion–exclusion gives

$$h_{a,N}(x) = \frac{1}{2^N (N-1)! \prod_{j=1}^{N} w_j} \sum_{S \subseteq \{1,\dots,N\}} (-1)^{|S|} \big(x + W_N - 2W_S\big)_+^{N-1}. \tag{9}$$

The knots are the signed sums $-W_N + 2W_S$.

Proposition 3.3 (Finite-product convergence)**.** *For the coupling using the same coordinates,*

$$|X_a - X_{a,N}| \leq \frac{a^{-N}}{a-1} \quad \textit{almost surely}, \qquad W_\infty(\mathcal{L}(X_a), \mathcal{L}(X_{a,N})) \leq \frac{a^{-N}}{a-1}. \tag{10}$$

Moreover, for every fixed $m \geq 0$, $h_{a,N} \to h_a$ in $C^m(\mathbb{R})$ as $N \to \infty$. On bounded t-sets,

$$|\Phi_a(t) - \Phi_{a,N}(t)| \leq \frac{t^2 a^{-2(N+1)}}{6(1-a^{-2})}. \tag{11}$$

Proof. The tail bound is the geometric series. The product estimate follows by telescoping and $|1 - \operatorname{sinc} z| \leq z^2/6$. For C^m convergence, choose $N_0 > m+2$. For $N \geq N_0$ both $\Phi_{a,N}$ and Φ_a are dominated by the integrable function $\prod_{j=1}^{N_0} |\operatorname{sinc}(ta^{-j})|$ after multiplication by $|t|^m$. Dominated convergence and Fourier inversion finish the proof. $\square$

4 Taylor coefficients, moments, Bernoulli numbers, and Bell polynomials

4.1 The source recurrence

Write

$$\Phi_a(t) = \sum_{k=0}^{\infty} c_{2k}(a) t^{2k}, \qquad c_0(a) = 1. \tag{12}$$

Odd coefficients vanish by symmetry. Comparing coefficients in (S3.28) gives the source recurrence

$$c_{2k}(a) = \frac{1}{a^{2k}-1} \sum_{\ell=0}^{k-1} \frac{(-1)^{k-\ell} c_{2\ell}(a)}{(2k-2\ell+1)!}, \qquad k \geq 1. \tag{S3.32}$$

The even moments

$$\mu_{2k}(a) = \int_{-b}^{b} x^{2k} h_a(x)\, \mathrm{d}x = (-1)^k (2k)! c_{2k}(a) \tag{S3.36–S3.37}$$

therefore satisfy an exact rational recurrence whenever a is rational or algebraic. The source iterates (S3.32) to obtain superfactorial decay estimates for $c_{2k}(a)$; these estimates underlie the very rapid endpoint algorithms in the dyadic case.

4.2 A closed cumulant formula

[N] The recurrence becomes more transparent after passing from moments to cumulants.

Theorem 4.1 (Bernoulli cumulants). *All odd cumulants of X_a vanish, and for $m \geq 1$,*

$$\boxed{\kappa_{2m}(X_a) = \frac{2^{2m} B_{2m}}{2m\,(a^{2m}-1)}.} \tag{13}$$

Consequently

$$\mu_n(a) = \mathcal{B}_n\big(0, \kappa_2, 0, \kappa_4, \ldots, \kappa_n\big), \tag{14}$$

where $\mathcal{B}_n$ is the complete exponential Bell polynomial.

Proof. For $U \sim \mathrm{Unif}[-1,1]$,

$$\log \mathbb{E}e^{zU} = \log \frac{\sinh z}{z} = \sum_{m\geq 1} \frac{2^{2m} B_{2m}}{2m(2m)!} z^{2m}.$$

Cumulants add under independence and scale homogeneously, hence

$$\kappa_{2m}(X_a) = \kappa_{2m}(U) \sum_{j\geq 1} a^{-2mj} = \frac{2^{2m} B_{2m}}{2m} \frac{1}{a^{2m}-1}.$$

The Bell-polynomial identity is the standard moment–cumulant relation. □

The first values are

$$\kappa_2 = \frac{1}{3(a^2-1)}, \qquad \kappa_4 = -\frac{2}{15(a^4-1)}, \qquad \kappa_6 = \frac{16}{63(a^6-1)}, \tag{15}$$

$$\mu_2 = \kappa_2, \qquad \mu_4 = 3\kappa_2^2 + \kappa_4, \qquad \mu_6 = 15\kappa_2^3 + 15\kappa_2\kappa_4 + \kappa_6. \tag{16}$$

This is computationally preferable to repeatedly expanding the infinite product.

4.3 q-Lambert structure

Let $q = a^{-2}$. Then

$$\frac{1}{a^{2m}-1} = \frac{q^m}{1-q^m}, \tag{17}$$

a classical Lambert-series factor. The logarithm of the characteristic function becomes

$$\log \Phi_a(t) = \sum_{m\geq 1} \frac{(-1)^m 2^{2m} B_{2m}}{2m(2m)!} \frac{q^m}{1-q^m} t^{2m}. \tag{18}$$

Expanding $q^m/(1-q^m) = \sum_{r\geq 1} q^{mr}$ gives a divisor-polynomial expansion

$$\log \Phi_a(t) = \sum_{n\geq 1} q^n \sum_{m|n} \frac{(-1)^m 2^{2m} B_{2m}}{2m(2m)!} t^{2m}. \tag{19}$$

This is the direct bridge from the general a family to the repository's q-series and q-binomial material. At $a = 2$, $q = 1/4$, while the half-base arithmetic of the bounded Fabius function introduces the complementary base $1/2$ through dyadic scaling.

5 The separated regime $a > 2$: smooth fractal splines

5.1 The central plateau and source polynomial pieces

Put

$$b = \frac{1}{a-1}, \qquad b_1 = b\left(1 - \frac{2}{a}\right) = \frac{a-2}{a(a-1)}. \tag{20}$$

For $a > 2$, the two intervals on which $h_a(ax \pm 1)$ can be nonzero are disjoint. Equation (S3.23) therefore gives $h_a'(x) = 0$ on $[-b_1, b_1]$. Using (6) and total mass 1 gives the source plateau

$$h_a(x) = \frac{a}{2} \qquad (|x| \le b_1). \tag{S3.40}$$

The source then observes recursively that h_a is linear on two intervals, quadratic on four intervals, and so forth. The union of all such polynomial intervals has full measure; the remaining null set is Cantor-like. We now identify this geometry exactly.

5.2 The singular attractor and its gaps

Define the contractions

$$S_-(x) = \frac{x-1}{a}, \qquad S_+(x) = \frac{x+1}{a}. \tag{21}$$

Definition 5.1 (Address Cantor set)**.** For $a > 2$, let

$$K_a = S_-(K_a) \cup S_+(K_a) = \left\{ \sum_{j=1}^{\infty} \varepsilon_j a^{-j} : \varepsilon_j \in \{-1, 1\} \right\}. \tag{22}$$

The central gap is $G_0 = (-b_1, b_1)$. For a word $w = (\varepsilon_1, \ldots, \varepsilon_n) \in \{-,+\}^n$, write $S_w = S_{\varepsilon_1} \circ \cdots \circ S_{\varepsilon_n}$ and $G_w = S_w(G_0)$.

Theorem 5.2 (Exact polynomial locus)**.** *For $a > 2$,*

$$[-b, b] \setminus K_a = \bigsqcup_{n \ge 0} \bigsqcup_{|w|=n} G_w. \tag{23}$$

On every generation-n gap G_w, the function h_a is a polynomial of exact degree n. Consequently the set of points admitting a neighborhood on which h_a is polynomial is precisely $[-b, b] \setminus K_a$.

Proof. The first-level images $S_\pm([-b, b])$ are disjoint and leave exactly the central gap G_0. Standard iteration of a separated two-map IFS gives (23).

Differentiate (S3.23) repeatedly. With

$$\delta_k = (-1)^{\mathrm{wt}_2(k-1)}, \qquad 1 \le k \le 2^n, \tag{24}$$

one obtains

$$h_a^{(n)}(x) = \left(\frac{a^2}{2}\right)^n a^{n(n-1)/2} \sum_{k=1}^{2^n} \delta_k \, h_a\left(a^n x + \sum_{j=1}^{n} a^{j-1}(-1)^{p_j(k-1)}\right). \tag{S3.49}$$

For x in a fixed generation-n gap, exactly one summand in (S3.49) lands in the central plateau and every other summand lands outside $[-b, b]$. Therefore $h_a^{(n)}$ is a nonzero constant on that gap. Differentiating once more gives zero there because the surviving plateau is constant. Hence the degree is exactly n.

Every point of K_a is accumulated by gaps of arbitrarily large generation. A single polynomial neighborhood of degree d cannot contain a generation-n gap with $n > d$. Thus no point of K_a belongs to the polynomial locus. □

Corollary 5.3 (Exact nonanalytic locus)**.** *Within its support, h_a is real analytic exactly on $[-b, b] \setminus K_a$, where it is locally a polynomial. It is nonanalytic at every point of K_a; on $\mathbb{R} \setminus [-b, b]$ it is identically zero and therefore analytic.*

Proof. The complement is polynomial by theorem 5.2. If h_a were analytic in a neighborhood of a point of K_a, that neighborhood would contain a gap. On the gap the analytic function agrees with a polynomial, so the identity theorem would make it that same polynomial throughout the neighborhood, contradicting theorem 5.2. □

This sharpens the source's statement that h_a is polynomial on a full-measure set and nonanalytic on a Cantor-type null set.

5.3 Breakpoints and exact moment evaluation

[S] The source describes the boundary points created after n differentiations as

$$x_0 = a^{-n} \left(\pm b + \sum_{k=0}^{n-1} \pm a^k \right). \tag{S3.50}$$

These are exactly the endpoints of the cylinder intervals $S_w([-b, b])$ and of their complementary gaps. At such points, the Taylor remainder formula based at $-b$,

$$h_a(x) = \frac{1}{n!} \int_{-b}^{x} h_a^{(n+1)}(t)(x - t)^n \, \mathrm{d}t, \tag{25}$$

combined with (S3.49), reduces values and derivatives to finite linear combinations of the even moments (S3.36–S3.37). The OCR of the source's fully indexed formulas (S3.54)–(S3.58) is too damaged to reproduce character by character, but their invariant content is exactly this finite moment reduction. A stable computational version is given in theorem 5.4.

Algorithm 5.4 (Exact polynomial on a prescribed gap)**.** Given $a > 2$ and a word w of length n:

1. compute the left endpoint x_w of G_w from the affine maps $S_\pm$;
2. compute moments $\mu_0, \mu_2, \dots, \mu_{2\lfloor (n-1)/2 \rfloor}$ by theorem 4.1;
3. use (25) and (S3.49) to evaluate $h_a^{(r)}(x_w)$ for $0 \le r \le n$;
4. return
$$P_w(x) = \sum_{r=0}^{n} \frac{h_a^{(r)}(x_w)}{r!} (x - x_w)^r.$$

   By theorem 5.2, $P_w = h_a$ throughout G_w.

5.4 Exact derivative norms

The source gives the exact dyadic norm law for up. The disjoint-support geometry shows that it extends to every $a \ge 2$.

Theorem 5.5 (Exact derivative norms)**.** *For every $a \ge 2$ and $n \ge 0$,*

$$\boxed{\|h_a^{(n)}\|_\infty = \frac{a^{(n+1)(n+2)/2}}{2^{n+1}}, \qquad \|h_a^{(n)}\|_1 = a^{n(n+1)/2}.} \tag{26}$$

Proof. The integral refinement gives $\|h_a\|_\infty \le a/2$, with equality on the plateau for $a > 2$ and at $x = 0$ for $a = 2$. The 2^n support intervals of the summands in (S3.49) are disjoint when $a > 2$ and meet only at endpoints when $a = 2$. Thus there is no cancellation in either the supremum or the absolute integral. The common coefficient is $(a^2/2)^n a^{n(n-1)/2}$. Multiplying it by $a/2$ gives the first formula. For the second, each rescaled copy has integral a^{-n}, so

$$\|h_a^{(n)}\|_1 = \left(\frac{a^2}{2}\right)^n a^{n(n-1)/2}\, 2^n a^{-n} = a^{n(n+1)/2}.$$

□

At $a = 2$, (26) reduces to the source identity

$$\|\operatorname{up}^{(n)}\|_\infty = \|\operatorname{up}^{(n)}\|_1 = 2^{n(n+1)/2}. \tag{S3.64}$$

6 Fractal geometry and the critical transition

6.1 Dimension and geometric zeta function

The central gap has length

$$\ell_0 = 2b_1 = \frac{2(a-2)}{a(a-1)}, \tag{27}$$

and the generation-n gaps have length $\ell_n = \ell_0 a^{-n}$ with multiplicity 2^n.

Theorem 6.1 (Dimension and complex dimensions). *For $a > 2$,*

$$\dim_H K_a = \dim_M K_a = D_a := \frac{\log 2}{\log a}. \tag{28}$$

The geometric zeta function of the complementary fractal string is

$$\zeta_{\mathcal{L}_a}(s) = \sum_{n\ge 0} 2^n \ell_n^s = \frac{\ell_0^s}{1-2a^{-s}}, \qquad \Re s > D_a. \tag{29}$$

It extends meromorphically to $\mathbb{C}$ with simple poles

$$s_k = D_a + \frac{2\pi i k}{\log a}, \qquad k \in \mathbb{Z}, \tag{30}$$

and

$$\operatorname*{res}_{s=s_k} \zeta_{\mathcal{L}_a}(s) = \frac{\ell_0^{s_k}}{\log a}. \tag{31}$$

Proof. The open-set condition holds because $a > 2$, and the similarity equation $2a^{-D} = 1$ gives (28) by Hutchinson's theorem [12]. The gap sum is geometric. Solving $1 - 2a^{-s} = 0$ gives the pole lattice, and differentiation of the denominator gives residue $\ell_0^{s_k}/\log a$. □

[R] The vertical arithmetic progression (30) is the canonical lattice pattern of a self-similar fractal string in the sense of Lapidus and van Frankenhuijsen [13, 14]. Here it enters the analysis of a C^∞ function: the singular set of the smooth spline carries the same complex dimensions as its complementary gap string.

6.2 An exact tube formula

For $\varepsilon > 0$, define the inner tube volume

$$V_a(\varepsilon) = |\{x \in [-b, b] \setminus K_a : \operatorname{dist}(x, K_a) < \varepsilon\}|. \tag{32}$$

Each gap of length ℓ contributes $\min(2\varepsilon, \ell)$.

Theorem 6.2 (Exact tube formula). *Let $a > 2$ and $0 < 2\varepsilon < \ell_0$. Put*

$$N(\varepsilon) = \left\lfloor \log_a \frac{\ell_0}{2\varepsilon} \right\rfloor. \tag{33}$$

Then

$$\boxed{V_a(\varepsilon) = 2\varepsilon\big(2^{N(\varepsilon)+1} - 1\big) + \frac{\ell_0(2/a)^{N(\varepsilon)+1}}{1 - 2/a}.} \tag{34}$$

If $x = \log_a(\ell_0/(2\varepsilon))$ and $\theta = \{x\}$, then

$$\varepsilon^{D_a - 1} V_a(\varepsilon) = \mathcal{M}_a(\theta) - 2\varepsilon^{D_a}, \tag{35}$$

where the one-periodic profile is

$$\mathcal{M}_a(\theta) = \ell_0^{D_a} \left[2^{2 - D_a - \theta} + \frac{2^{1 - D_a} a^{(D_a - 1)(1 - \theta)}}{1 - 2/a} \right], \qquad 0 \le \theta < 1. \tag{36}$$

The profile is continuous under periodic identification and nonconstant. Hence K_a is not Minkowski measurable, although its upper and lower Minkowski contents are finite and positive.

Proof. The gaps with $n \le N$ are wider than 2ε and contribute 2ε each; the remaining gaps are completely covered. Summing the finite geometric count and the infinite geometric length tail gives (34). Substitution of $\varepsilon = (\ell_0/2)a^{-x}$ and $a^{D_a} = 2$ gives (35)–(36). The two positive exponentials in θ cannot sum to a constant for $a > 2$. □

The logarithmic average of the oscillatory content is explicit:

$$\int_0^1 \mathcal{M}_a(\theta)\, \mathrm{d}\theta = \ell_0^{D_a} 2^{1 - D_a} \left(\frac{1}{\log 2} + \frac{1}{\log(a/2)} \right). \tag{37}$$

6.3 The local polynomial degree as a random variable

Choose a point uniformly with respect to normalized Lebesgue measure on $[-b, b]$. Since K_a has measure zero, the point lies in a unique gap. Let N_a be the exact degree of the polynomial piece containing it.

Theorem 6.3 (Geometric degree law). *For $a > 2$,*

$$\mathbb{P}(N_a = n) = \frac{a - 2}{a} \left(\frac{2}{a} \right)^n, \qquad n = 0, 1, 2, \ldots. \tag{38}$$

Therefore

$$\mathbb{E} N_a = \frac{2}{a - 2}, \qquad \operatorname{Var}(N_a) = \frac{2a}{(a - 2)^2}. \tag{39}$$

Let $p_a = (a - 2)/a$. As $a \downarrow 2$,

$$p_a N_a \xrightarrow{d} \operatorname{Exp}(1), \qquad \frac{a - 2}{2} N_a \xrightarrow{d} \operatorname{Exp}(1). \tag{40}$$

Proof. The total length of all generation-n gaps is $2^n \ell_0 a^{-n}$. Dividing by $2b = 2/(a-1)$ gives (38). The moment formulas are those of a geometric law on $\mathbb{N}_0$. Finally,

$$\mathbb{P}(p_a N_a > x) = (1 - p_a)^{\lfloor x/p_a \rfloor + 1} \longrightarrow \mathrm{e}^{-x}.$$

□

This theorem makes the phrase "polynomial almost everywhere" quantitative. Near the critical base, a typical polynomial piece has a very large degree, of order $(a-2)^{-1}$.

6.4 The dyadic phase transition

As $a \downarrow 2$,

$$D_a = 1 - \frac{a-2}{2\log 2} + \mathcal{O}((a-2)^2), \qquad \ell_0 = (a-2) + \mathcal{O}((a-2)^2). \tag{41}$$

Thus the null singular set thickens to dimension 1, the central plateau collapses, and the mean local degree diverges. At $a = 2$ the two IFS pieces merely touch; the complement has no open gaps at all. The limit function is nowhere analytic on its support, as the source states. This is a genuine regularity phase transition, not merely a change in support geometry.

7 The critical member $h_2 = \mathrm{up}$

7.1 Fourier product, equation, support, and derivatives

[S] At $a = 2$,

$$\mathrm{up}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \mathrm{e}^{-\mathrm{i}tx} \prod_{k=1}^{\infty} \mathrm{sinc}(t 2^{-k})\, \mathrm{d}t, \tag{S3.60}$$

$$\mathrm{up}'(x) = 2\,\mathrm{up}(2x+1) - 2\,\mathrm{up}(2x-1), \qquad \mathrm{supp}\,\mathrm{up} = [-1, 1]. \tag{S3.61–S3.62}$$

Since

$$\sum_{j=1}^{n} 2^{j-1} (-1)^{p_j(k-1)} = 2^n + 1 - 2k,$$

formula (S3.49) becomes

$$\mathrm{up}^{(n)}(x) = 2^{n(n+1)/2} \sum_{k=1}^{2^n} \delta_k\, \mathrm{up}(2^n x + 2^n + 1 - 2k). \tag{S3.63}$$

The exact norm identity is the $a = 2$ case of theorem 5.5.

7.2 Taylor polynomials and nowhere analyticity

[S] At a dyadic rational represented at depth n, the Taylor series of up terminates after degree at most n; at every nondyadic point of $[-1, 1]$, the Taylor series has radius of convergence zero. Hence up is nonanalytic at every point of its support. The dyadic points are not analytic exceptions: their finite Taylor polynomials do not agree with up on any neighborhood. This is the critical counterpart of theorem 5.3: for $a > 2$ the polynomial Taylor series is locally exact off K_a, while for $a = 2$ there are no open polynomial gaps.

7.3 Moment formulas and exact dyadic values

Let

$$\mu_{2r} = \int_{-1}^{1} x^{2r}\,\mathrm{up}(x)\,\mathrm{d}x, \qquad m_p^+ = \int_0^1 x^p\,\mathrm{up}(x)\,\mathrm{d}x. \tag{42}$$

Then

$$\mu_{2r} = (-1)^r (2r)!\,c_{2r}(2), \tag{S3.65}$$

and integration by parts in (S3.61–S3.62) gives, for $p \geq 0$,

$$m_p^+ = \frac{1}{(p+1)2^{p+1}} \sum_{r=0}^{\lfloor (p+1)/2 \rfloor} \binom{p+1}{2r} \mu_{2r}. \tag{43}$$

The especially useful endpoint value is

$$\mathrm{up}(-1+2^{-n}) = \frac{m_{n-1}^+}{2^{n(n-1)/2}(n-1)!}, \qquad n \geq 1. \tag{S3.68}$$

More generally, for $1 \leq k \leq 2^n$,

$$\begin{aligned} \mathrm{up}(-1+k2^{-n}) &= \frac{2^{-n(n-1)/2}}{n!} \sum_{j=1}^{k} \delta_j \int_{-1}^{1} \mathrm{up}(t) \left(k-j+\frac{1}{2}-\frac{t}{2}\right)^n \mathrm{d}t \\ &= \frac{2^{-n(n-1)/2}}{n!} \sum_{j=1}^{k} \delta_j \sum_{r=0}^{\lfloor n/2 \rfloor} \binom{n}{2r} \left(k-j+\frac{1}{2}\right)^{n-2r} 2^{-2r}\mu_{2r}. \end{aligned} \tag{44}$$

This is a cleaned version of the source's formula (S3.67).

7.4 Binary recursion and a rapidly convergent evaluation scheme

For $n \geq 1$, define

$$\varphi_n(x) = \mathrm{up}(x) + \mathrm{up}(x - 2^{-n}). \tag{45}$$

On $I_n = [-1+2^{-n}, -1+2^{-n+1}]$, the source observes that $\varphi_n^{(n+1)} = 0$, because the first two Thue–Morse signs cancel. Therefore

$$\varphi_n(x) = \sum_{r=0}^{n} \frac{\mathrm{up}^{(r)}(-1+2^{-n})}{r!}(x+1-2^{-n})^r, \qquad x \in I_n, \tag{S3.70}$$

and

$$\mathrm{up}(x) = \sum_{r=0}^{n} \frac{\mathrm{up}^{(r)}(-1+2^{-n})}{r!}(x+1-2^{-n})^r - \mathrm{up}(x-2^{-n}). \tag{S3.71}$$

Iterating (S3.71) according to the binary digits of $x+1$ produces the source's series (S3.73). The exact printed indices are partly corrupted in the OCR, but the algorithm and its error bound are unambiguous:

$$|R_N(x)| \leq \mathrm{up}(-1+2^{-N}), \tag{46}$$

and the right side decreases faster than any geometric sequence because of (S3.68) and the superfactorial moment bounds. This is the conceptual ancestor of modern exact dyadic evaluators in the repository and in [8].

8 Thue–Morse signs and polynomial reproduction

8.1 The derivative operator is a Prouhet filter

The signs in the source recurrence

$$\delta_{2k-1} = \delta_k, \qquad \delta_{2k} = -\delta_k, \qquad \delta_1 = 1, \tag{47}$$

are exactly $\delta_k = (-1)^{\mathrm{wt}_2(k-1)}$. Their generating polynomial is

$$\sum_{k=0}^{2^n-1} (-1)^{\mathrm{wt}_2(k)} z^k = \prod_{j=0}^{n-1} (1 - z^{2^j}). \tag{48}$$

Thus the nth derivative formula is the product of n two-branch difference filters. The zero of order n at $z = 1$ gives the Prouhet cancellation

$$\sum_{k=0}^{2^n-1} (-1)^{\mathrm{wt}_2(k)} k^r = 0, \qquad 0 \le r < n. \tag{49}$$

This is why high-order polynomial pieces and exact dyadic identities emerge from what first looks like a functional-differential equation.

8.2 Characteristic polynomial for translate coefficients

[S] The source considers locally finite sums

$$\phi(x) = \sum_{k\in\mathbb{Z}} c_k \,\mathrm{up}(x - k2^{-n}). \tag{S3.76}$$

For ϕ to be a polynomial of degree at most n, its $(n+1)$st derivative must vanish, which is equivalent to the constant-coefficient difference system

$$\sum_{j=1}^{2^{n+1}} c_{k+j}\delta_j = 0, \qquad k \in \mathbb{Z}. \tag{S3.77}$$

The characteristic polynomial is

$$Q_n(z) = \sum_{k=1}^{2^{n+1}} \delta_k z^{2^{n+1}-k}. \tag{S3.79}$$

The source recurrence

$$Q_n(z) = Q_{n-1}(z)(z^{2^n} - 1), \qquad Q_0(z) = z - 1, \tag{S3.80–S3.81}$$

gives

$$\boxed{Q_n(z) = \prod_{j=0}^{n} (z^{2^j} - 1).} \tag{50}$$

A root of exact order 2^m has multiplicity $n - m + 1$ for $1 \le m \le n$, while 1 has multiplicity $n + 1$. Hence every solution of (S3.77) is a finite sum of polynomial–exponential sequences

$$c_k = \sum_{\omega : Q_n(\omega)=0} \left(\sum_{r=0}^{\mathrm{mult}(\omega)-1} A_{\omega,r} k^r \right) \omega^k. \tag{51}$$

The high multiplicity of 1 is the polynomial-reproduction mechanism; the other roots of unity describe oscillatory null modes.

8.3 Uniqueness of the universal representing function

[S] The source proves a uniqueness theorem strengthening earlier Rvachev results. A clean statement is as follows.

Theorem 8.1 (Source uniqueness principle). *Let $\varphi \in C_c^\infty(\mathbb{R})$ have support $[-1,1]$ and mass 1. Assume*

$$\lim_{n\to\infty} \frac{\|\varphi^{(n)}\|_\infty}{2^{(n+1)(n+2)/2}} = 0, \tag{52}$$

and assume that for every n there are coefficients $c_k^{(n)}$ such that the locally finite sum

$$\sum_{k\in\mathbb{Z}} c_k^{(n)} \varphi(x - k2^{-n}) = x^n. \tag{53}$$

Then $\varphi = \mathrm{up}$.

The proof builds a sequence φ_n satisfying the same mass and reproduction conditions and

$$\varphi_n'(x) = 2\varphi_{n+1}(2x+1) - 2\varphi_{n+1}(2x-1). \tag{54}$$

Comparing with the identical integral recursion for up, the derivative-growth hypothesis forces the successive differences to contract to zero. The growth condition is sharp in the source's sense: $\mathrm{up} + \alpha\,\mathrm{up}'$ retains mass and polynomial reproduction but violates (52) with limiting ratio $|\alpha|$.

9 The Fup construction and a natural hierarchy

9.1 The surviving source definition

[S] The attached source ends shortly after using

$$\mathrm{sinc}(t/2) = \cos(t/4)\,\mathrm{sinc}(t/4) \tag{55}$$

to rewrite

$$\widehat{\mathrm{up}}(t) = \cos(t/4)\,(\mathrm{sinc}(t/4))^2 \prod_{j=3}^{\infty} \mathrm{sinc}(t2^{-j}). \tag{S3.98}$$

It defines Fup_1 by

$$\widehat{\mathrm{Fup}_1}(t) = (\mathrm{sinc}(t/4))^2 \prod_{j=3}^{\infty} \mathrm{sinc}(t2^{-j}), \tag{S3.99}$$

and obtains

$$\mathrm{up}(x) = \frac{1}{2}\left(\mathrm{Fup}_1(x - 1/4) + \mathrm{Fup}_1(x + 1/4)\right). \tag{S3.100}$$

No later pages of the Fup_n section are present in the attachment.

9.2 A canonical continuation

[N] There is a simple hierarchy forced by the same factorization. Put

$$H(t) = \prod_{j=2}^{\infty} \mathrm{sinc}(t2^{-j}) \tag{56}$$

and define G_n for $n \geq 0$ by

$$\widehat{G}_n(t) = H(t)\,\mathrm{sinc}(t2^{-(n+1)}). \tag{57}$$

Then $G_0 = \mathrm{up}$ and $G_1 = \mathrm{Fup}_1$.

Proposition 9.1 (Dyadic Fup ladder). *For $n \geq 1$,*

$$G_{n-1}(x) = \frac{1}{2}\left(G_n(x - 2^{-(n+1)}) + G_n(x + 2^{-(n+1)})\right). \tag{58}$$

Moreover,

$$G_n \longrightarrow 2\,\mathrm{up}(2\,\cdot) \quad \textit{in } C^m(\mathbb{R}) \textit{ for every fixed } m. \tag{59}$$

The support radius of G_n is $1/2 + 2^{-(n+1)}$.

Proof. Identity (55) gives $\widehat{G}_{n-1}(t) = \cos(t2^{-(n+1)})\widehat{G}_n(t)$, which is (58). Pointwise, $\widehat{G}_n(t) \to H(t)$, and H is the Fourier transform of the density of $\sum_{j\geq 2} 2^{-j}U_j = X_2/2$, namely $2\,\mathrm{up}(2x)$. Since $|\widehat{G}_n - H| \leq 2|H|$ and $|t|^m H(t) \in L^1$, dominated Fourier inversion gives C^m convergence. The support radius is the sum of the uniform half-widths in (57). □

Iterating (58) expresses up as an average of 2^n translates of G_n. In the limit, the translate locations become the usual centered dyadic Bernoulli sum, whose law is uniform on $[-1/2, 1/2]$. The limit recovers the first-step convolution identity for up. This makes the Fup construction a finite Rademacher factorization of the uniform smoothing mask.

10 Parameter limits: Gaussian, critical, and uniform

10.1 The many-small-summands limit $a \downarrow 1$

The variance is

$$\sigma_a^2 = \mathrm{Var}(X_a) = \frac{1}{3(a^2-1)}. \tag{60}$$

Let $Z_a = X_a/\sigma_a$.

Theorem 10.1 (Gaussian limit and first characteristic correction). *As $a \downarrow 1$,*

$$Z_a \xrightarrow{d} N(0,1). \tag{61}$$

For every fixed $T > 0$, uniformly for $|t| \leq T$,

$$\mathbb{E}\mathrm{e}^{\mathrm{i}tZ_a} = \mathrm{e}^{-t^2/2}\left[1 - \frac{a^2-1}{20(a^2+1)}t^4 + \mathcal{O}_T((a-1)^2)\right]. \tag{62}$$

More generally, the standardized $2m$th cumulant is

$$\lambda_{2m}(a) = \frac{3^m 2^{2m} B_{2m}(a^2-1)^m}{2m(a^{2m}-1)} = \mathcal{O}((a-1)^{m-1}), \qquad m \geq 2. \tag{63}$$

Proof. The triangular-array coefficients in $Z_a = \sum_j \sqrt{3(a^2-1)}a^{-j}U_j$ have maximal magnitude tending to zero and total variance one, so the Lindeberg theorem applies. Alternatively, insert (63) in the logarithm of the characteristic function. The exact fourth standardized cumulant is

$$\lambda_4(a) = -\frac{6}{5}\frac{a^2-1}{a^2+1}. \tag{64}$$

All higher contributions to the logarithm are $\mathcal{O}_T((a-1)^2)$, and exponentiation gives (62). □

[C] A natural next theorem is a uniform local Edgeworth expansion for the normalized densities and their derivatives. The characteristic expansion and superpolynomial Fourier decay make this plausible, but uniform control as $a \downarrow 1$ requires a careful split between central and large frequencies.

10.2 The single-dominant-summand limit $a \to \infty$

Put $Y_a = aX_a$. Then

$$Y_a = U_1 + T_a, \qquad |T_a| \le \frac{1}{a-1}. \tag{65}$$

Theorem 10.2 (Uniform limit with quantitative bounds)**.** *Let $U \sim \mathrm{Unif}[-1,1]$. For $a > 2$,*

$$W_\infty(\mathcal{L}(Y_a), \mathcal{L}(U)) \le \frac{1}{a-1}, \qquad d_{\mathrm{TV}}(\mathcal{L}(Y_a), \mathcal{L}(U)) \le \frac{1}{2(a-1)}. \tag{66}$$

If g_a is the density of Y_a, then

$$g_a(y) = \frac{1}{2} \quad \textit{for} \quad |y| \le \frac{a-2}{a-1}. \tag{67}$$

Thus the convergence is exact on an expanding central interval; all discrepancy is confined to two boundary layers, each of width $2/(a-1)$ (total width $4/(a-1)$).

Proof. The coupling (65) gives the Wasserstein bound. Conditional on $T_a = r$, the law is uniform on $[r-1, r+1]$, whose total variation distance from $[-1,1]$ is $|r|/2$; convexity of total variation gives the second bound. Finally, $g_a(y) = h_a(y/a)/a$, and the plateau (S3.40) becomes (67). □

11 Exact negative-Laplace renormalization in arbitrary base

11.1 Endpoint shift and product

Let

$$Y_a = X_a + b \in [0, 2b], \qquad f_a(y) = h_a(-b + y). \tag{68}$$

Since $U_j + 1$ is uniform on $[0,2]$,

$$\int_0^{2b} \mathrm{e}^{-sy} f_a(y)\,\mathrm{d}y = \prod_{j=1}^{\infty} \frac{1 - \mathrm{e}^{-2sa^{-j}}}{2sa^{-j}}. \tag{69}$$

Define, for $u > 0$,

$$\kappa(u) = \log \frac{1 - \mathrm{e}^{-u}}{u}, \qquad \Lambda_a(u) = \sum_{j=1}^{\infty} \kappa(ua^{-j}). \tag{70}$$

Then the logarithm of (69) is $\Lambda_a(2s)$, and

$$\Lambda_a(au) = \Lambda_a(u) + \kappa(u). \tag{71}$$

11.2 The exact periodic decomposition

Put $L_a = \log a$ and, for $x \in \mathbb{R}$,

$$R_a(x) = \sum_{n=0}^{\infty} \log\bigl(1 - \mathrm{e}^{-a^{x+n}}\bigr), \tag{72}$$

which is negative and exponentially convergent. Define

$$P_a(x) = \Lambda_a(a^x) + \frac{L_a}{2}x^2 - \frac{L_a}{2}x + R_a(x). \tag{73}$$

Theorem 11.1 (General-base exact periodic decomposition)**.** *For every $a > 1$, P_a is one-periodic. For $u > 0$,*

$$\boxed{\Lambda_a(u) = -\frac{(\log u)^2}{2L_a} + \frac{\log u}{2} + P_a(\log_a u) + E_a(u),} \tag{74}$$

where

$$E_a(u) = -R_a(\log_a u) = \sum_{n=0}^{\infty} -\log(1 - \mathrm{e}^{-ua^n}) \geq 0. \tag{75}$$

Moreover,

$$0 \leq E_a(u) \leq \frac{\mathrm{e}^{-u}}{(1 - \mathrm{e}^{-u})(1 - \mathrm{e}^{-(a-1)u})}. \tag{76}$$

In particular $E_a(u) = \mathcal{O}_a(\mathrm{e}^{-u})$ as $u \to \infty$.

Proof. Using (71) and $R_a(x+1) = R_a(x) - \log(1 - \mathrm{e}^{-a^x})$, one obtains

$$P_a(x+1) - P_a(x) = \kappa(a^x) + L_a x + R_a(x+1) - R_a(x) = 0.$$

Rearranging (73) gives (74). For the tail, use $-\log(1-z) \leq z/(1-z)$, $a^n \geq 1 + n(a-1)$, and sum the resulting geometric series. □

For $a = 2$, (76) becomes $E_2(u) \leq \mathrm{e}^{-u}/(1 - \mathrm{e}^{-u})^2$, exactly the dyadic form appearing in the primary repository exposition. Thus theorem 11.1 is a base-a renormalization of the same mechanism rather than an unrelated analogy.

11.3 Gamma–zeta Fourier coefficients

Let

$$\overline{P}_a = \int_0^1 P_a(x)\,\mathrm{d}x, \qquad \Psi_a(x) = P_a(x) - \overline{P}_a, \qquad \chi_k = \frac{2\pi \mathrm{i} k}{L_a}. \tag{77}$$

Theorem 11.2 (General-base Gamma–zeta modes)**.** *For $k \neq 0$,*

$$\boxed{\widehat{\Psi}_a(k) = \int_0^1 \Psi_a(x) \mathrm{e}^{-2\pi \mathrm{i} kx}\,\mathrm{d}x = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L_a}.} \tag{78}$$

The Fourier series is absolutely convergent with all derivatives; in fact the coefficients decay exponentially in $|k|$. Therefore P_a is real analytic and nonconstant.

Proof architecture. In the strip $-1 < \Re s < 0$, integration by parts gives the Mellin transform

$$\int_0^\infty \kappa(u) u^{s-1}\,\mathrm{d}u = -\Gamma(s)\zeta(s+1). \tag{79}$$

Mellin summation of the dilation equation samples this transform at $s = -\chi_k$, producing (78); the polynomial terms in (74) affect only the zero mode. Stirling's formula for $\Gamma(\sigma + \mathrm{i}t)$ gives exponential decay. Nonconstancy follows already from the nonzero first mode; more generally Γ has no zeros and the classical zero-free line $\zeta(1 + \mathrm{i}t) \neq 0$ shows that every nonzero mode is nonzero. □

[R] Formula (78) is the arbitrary-base counterpart of the exact dyadic periodic correction in the repository. It also belongs to the general Mellin theory of harmonic sums developed by Flajolet, Gourdon, and Dumas [16]. The same "lattice in logarithmic scale" mechanism appears independently in theorem 6.1; one periodicity comes from endpoint Laplace renormalization, the other from the geometric gap string.

11.4 Lower Lambert coordinate and the endpoint program

Ignoring the bounded periodic derivative and exponentially small tail in the saddle equation for (69), put

$$r_a(y)a^{-r_a(y)} = \frac{y\sqrt{a}}{2}. \tag{80}$$

The large solution is

$$r_a(y) = -\frac{1}{\log a} W_{-1}\left(-\frac{y\sqrt{a}\log a}{2}\right), \qquad y \downarrow 0. \tag{81}$$

This is the natural general-base lower-Lambert coordinate. The exact saddle is shifted by a bounded periodic term involving P'_a.

Conjecture 11.3 (All-orders general-base endpoint expansion)**.** *Let* $f_a(y) = h_a(-b+y)$*. For fixed* $a > 1$*, the exact positive saddle* $s_a(y)$ *defined by*

$$y + 2\Lambda'_a(2s_a(y)) = 0 \tag{82}$$

exists and is unique for sufficiently small $y > 0$*. There are real-analytic one-periodic functions* $A_{a,j}$ *such that the saddle expansion*

$$f_a(y) \sim \frac{\exp\big(s_a(y)y + \Lambda_a(2s_a(y))\big)}{\sqrt{8\pi\Lambda''_a(2s_a(y))}}\left(1 + \sum_{j\geq 1}\frac{A_{a,j}(\log_a(2s_a(y)))}{r_a(y)^j}\right) \tag{83}$$

holds to arbitrary order. Equivalently, $\log f_a(y)$ *has a transseries whose leading scale is* $-(\log a)r_a(y)^2/2$*, whose next terms include a linear term and* $-\frac{1}{2}\log r_a(y)$*, and whose order-one oscillation is* $P_a(r_a(y)+\frac{1}{2})$ *after normalization.*

At $a = 2$, normalization and a half-unit phase shift reduce this program to the exact lower-W_{-1} and Gamma–zeta hierarchy already developed in the repository. The point of theorem 11.3 is not to rederive that dyadic case, but to identify the correct general-base structure and the periodic term that any sharp formula must retain.

12 Connections to existing theories

12.1 Infinite convolutions and smoothing transforms

The probability law of X_a is an infinite convolution of compactly supported absolutely continuous measures. Unlike Bernoulli convolutions, whose regularity may depend delicately on arithmetic properties of the contraction, every digit here is uniform and each convolution step adds smoothing. Nevertheless, the *address geometry* of the separated regime is the same two-map Cantor geometry. This separation between a smooth full-support density and a fractal nonanalytic locus is one of the distinctive features of h_a.

12.2 Refinement equations, subdivision, and wavelets

The Fourier equation $\Phi_a(t) = \mathrm{sinc}(t/a)\Phi_a(t/a)$ is a refinement relation with an analytic, nonfinite mask. The finite approximants $h_{a,N}$ are box splines, while the limit is infinitely smooth. Daubechies–Lagarias refinement theory emphasizes finite masks, matrix products, and Hölder regularity [10, 11]; the present family supplies a complementary exactly solvable model in which local regularity is not merely estimated but encoded by a geometric gap address.

12.3 Thue–Morse, Prouhet cancellation, and roots of unity

The derivative signs, dyadic value formulas, and polynomial-reproduction recurrence are all manifestations of the same factorization (48). The source's characteristic polynomial (50) records not only the zero at 1 responsible for Prouhet cancellation but the full hierarchy of 2-power roots of unity and their multiplicities. This explains the simultaneous appearance of polynomial coefficients, trigonometric sequences, and 2-adic valuations in Rvachev's representation theorem.

12.4 Bernoulli and Bell polynomials

Bernoulli numbers enter through the logarithm of $\sinh z/z$; Bell polynomials reconstruct moments from cumulants. In the dyadic corpus, Bernoulli and Bell polynomials also arise in Euler–Maclaurin and endpoint expansions. Here their role is more elementary and more general: they give a closed moment calculus for every $a > 1$ before any dyadic specialization.

12.5 Two distinct logarithmic lattices

Two periodic phenomena occur in this report.

1. The gap string has complex dimensions $D_a + 2\pi \mathrm{i}k/\log a$, producing tube oscillations in $\log_a(1/\varepsilon)$.
2. The negative-Laplace equation has Fourier modes indexed by the same vertical spacing, producing endpoint oscillations in $\log_a u$.

They have different amplitudes and arise from different objects, but both are consequences of an exact multiplicative lattice. A promising future problem is to determine whether there is a direct transform carrying one periodic profile into the other.

13 Numerical experiments and reproducibility

The accompanying script `experiments.py` uses only NumPy, Matplotlib, and mpmath. It creates all figures and CSV tables in the archive. Random experiments use the fixed seed `20260828`; the omitted random-series tail is deterministically below 10^{-12}.

13.1 Gap hierarchy and exact polynomial degree

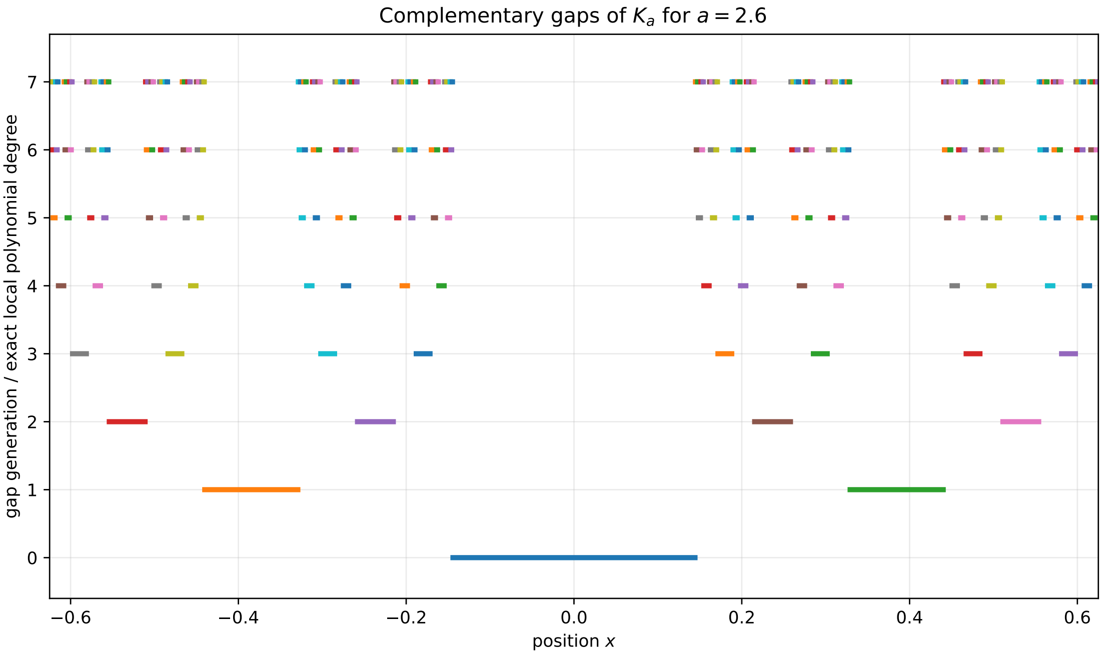

Figure 1: The gaps G_w for $a = 2.6$. Vertical level n is both the gap generation and, by theorem 5.2, the exact degree of h_a on that interval.

13.2 Tube oscillation

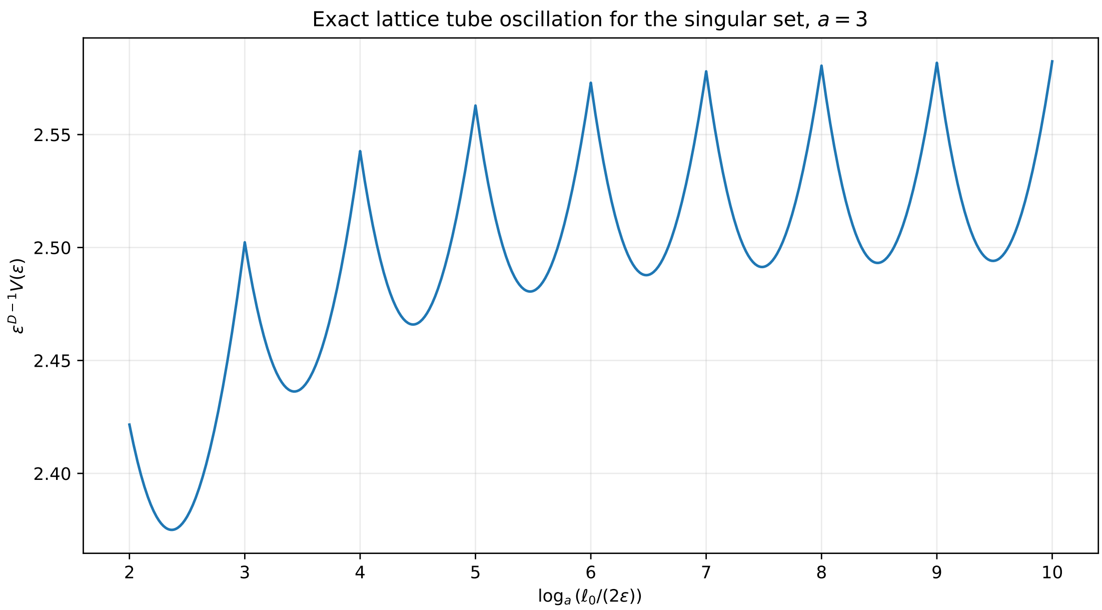

Figure 2: The exact normalized tube volume for $a = 3$. The approach to a nonconstant one-periodic profile verifies the lattice oscillation in theorem 6.2.

13.3 Critical local-degree limit

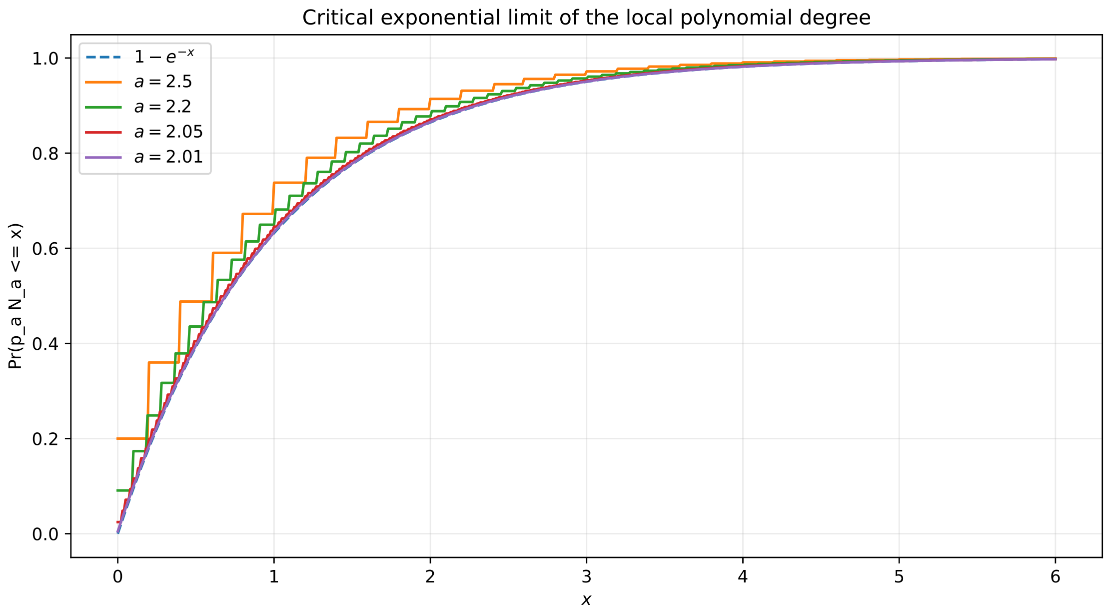

Figure 3: Distribution functions of $p_a N_a$ approaching $1 - \mathrm{e}^{-x}$ as $a \downarrow 2$.

13.4 General-base periodic correction

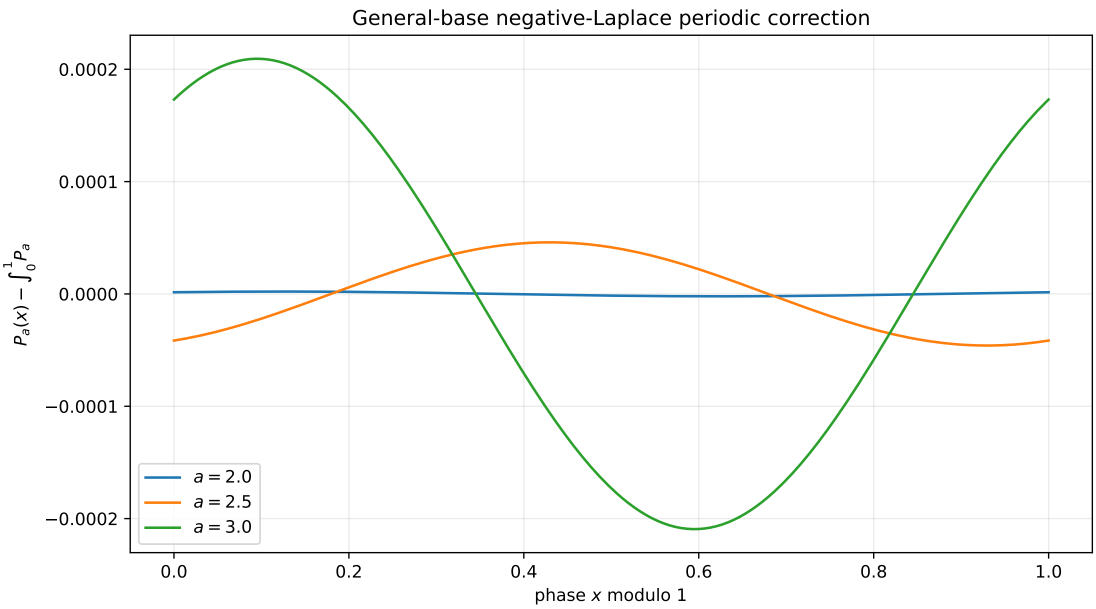

Figure 4: Centered periodic corrections $P_a - \overline{P}_a$. The dyadic oscillation is small but nonzero; its amplitude grows substantially for the displayed larger bases.

On a grid, the maximum residual in $P_a(x+1) - P_a(x)$ is below 3×10^{-14} for $a \in \{1.2, 1.5, 2, 2.6, 3, 5\}$. Numerical Fourier coefficients agree with (78) to between 10^{-12} and 10^{-16} in the supplied table; the larger dyadic first-mode error reflects resolving a Fourier mode of amplitude about 10^{-6} with double-precision quadrature, not a visible failure of the identity.

13.5 Gaussian cumulant scaling

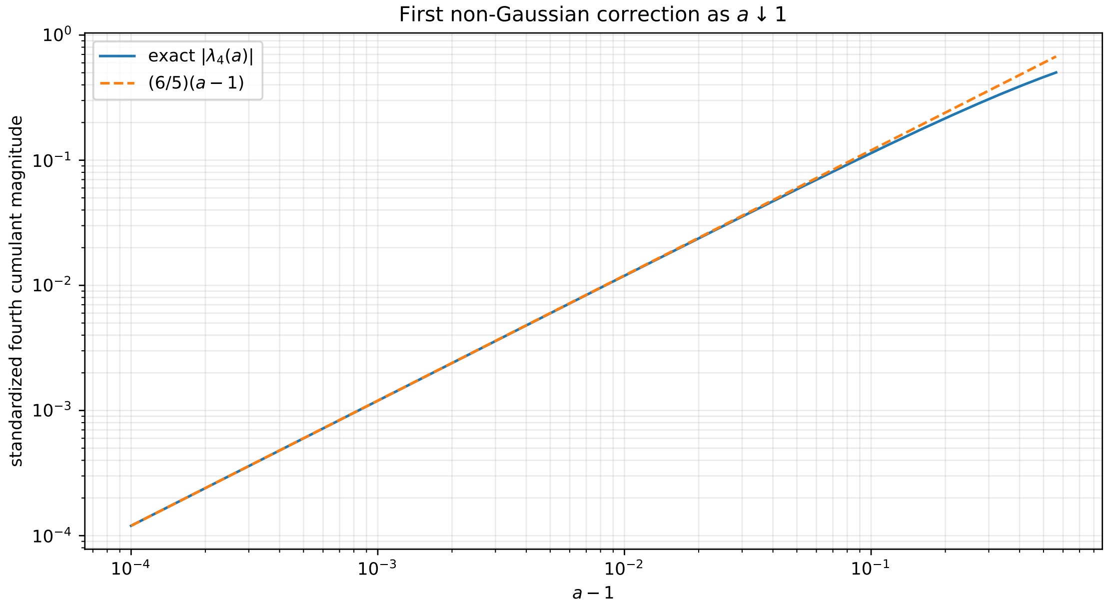

Figure 5: The exact standardized fourth cumulant and its first asymptotic term.

For $a = 2.6$, a fixed-seed simulation of 250,000 random sums confirms the exact Bell moments through order 10 at the expected Monte Carlo accuracy. Exact values and errors are stored in `data/moment_validation_a_2_6.csv`.

14 Conjectures and future research directions

Conjecture 14.1 (Overlap-regime nowhere analyticity)**.** *For every $1 < a \leq 2$, h_a is nonanalytic at every point of the interior of its support.*

The case $a = 2$ is the source theorem. For $1 < a < 2$ the two refinement branches overlap, so the gap-degree argument disappears. A proof may require a local Fourier or quasi-analyticity criterion adapted to the dilation equation.

Conjecture 14.2 (Critical profile of polynomial pieces)**.** *There is a nontrivial scaling limit for the polynomial containing a Lebesgue-typical point when $a = 2 + \eta$, after rescaling its degree by $\eta/2$, its interval by its random gap length, and its amplitude by the exact derivative norm in theorem 5.5.*

The exponential degree law identifies the first marginal. The remaining problem is a joint limit of word addresses, polynomial coefficients, and moments as the gaps collapse.

Conjecture 14.3 (All-order general-base saddle expansion)**.** *The formal saddle program in theorem 11.3 is valid uniformly for a in compact subsets of $(1, \infty)$, with coefficients generated by universal Bell polynomials in the derivatives of P_a.*

This would connect the repository's dyadic lower-Lambert coefficients to a continuous base parameter and make Bernoulli–Bell structure appear at both the moment and endpoint levels.

Conjecture 14.4 (Arithmetic of algebraic breakpoints)**.** *For algebraic $a > 2$, exact values of h_a and its derivatives at cylinder endpoints belong to the field generated by a and the even moments, with denominator growth controlled by the ideals generated by $a^{2m} - 1$.*

The source already proves rational dependence on a at the formal level. The cumulant formula suggests a systematic algebraic-denominator theory generalizing the dyadic rational arithmetic.

Conjecture 14.5 (A bridge between the two periodic profiles)**.** *There exists a natural spectral operator, built from the smoothing transform or its transfer operator, relating the tube profile $\mathcal{M}_a$ of* (36) *to the negative-Laplace profile P_a of* (73)*.*

The two profiles have the same logarithmic period but very different Fourier amplitudes. Even a rigorous obstruction to a simple relation would clarify how geometry and endpoint probability decouple.

Concrete research program

The following projects appear especially promising.

1. Formalize the $a > 2$ IFS, exact polynomial locus, derivative norms, degree law, and tube formula in Lean, using the repository's existing weighted-uniform distribution API.
2. Develop a transfer-operator proof or disproof of theorem 14.1.
3. Derive the first two general-base saddle coefficients explicitly and check their specialization against the formal dyadic coefficients.
4. Study total positivity, variation diminution, and approximation order of the finite box splines $h_{a,N}$ and their limit.
5. Extend the two-branch family to multiple uniform digits and nongeometric scales. The separated singular locus then becomes a graph-directed or self-affine set, while the Fourier product remains explicit.
6. Complete the historical Fup_n translation from a full source copy and compare it with the canonical hierarchy of theorem 9.1.
7. Investigate stable algorithms for exact polynomial evaluation on deep gaps and for certified endpoint inversion via W_{-1}.
8. Quantify the convergence $h_{a,N} \to h_a$ in Sobolev, Besov, and weighted Fourier norms, and compare with subdivision regularity bounds.
9. Connect the roots-of-unity null modes in (51) to wavelet vanishing moments and multiresolution decompositions generated by up.

15 Conclusion

The attached chapter contains substantially more than the isolated definition of the Rvachev bump. It presents a general Fourier-product existence mechanism, a derivative ladder, the one-parameter density h_a, a probability model, superfast moment recurrences, full-measure polynomial splines for $a > 2$, the critical nowhere-analytic function up, exact dyadic values, polynomial reproduction, and a uniqueness principle. In modern terms, it sits at the intersection of infinite convolution, refinement equations, box splines, automatic sequences, and compactly supported smooth analysis.

The general parameter a reveals a particularly clean frontier. For $a > 2$, the smooth function is locally polynomial off an exact Cantor set, and every geometric statistic of that set—dimension, complex dimensions, tube volume, and degree distribution—has a closed form. At $a = 2$ the gaps close, the typical degree diverges, and nowhere analyticity takes over. At the opposite parameter extremes, the law becomes Gaussian after normalization as $a \downarrow 1$ and uniform after scaling as $a \to \infty$. Finally, the negative-Laplace renormalization and Gamma–zeta periodic correction survive in every base, providing the correct setting for a general lower-Lambert endpoint theory.

A Repository corpus map and nonduplication protocol

The repository tree consulted on 28 August 2026 separates living documents, temporary frontier inboxes, vendored literature, and frozen historical versions. The following map records how those layers were used.

Corpus layer	Representative contents	Role in this report
Living reference volumes	Glossary; primary Fabius–Rvachev exposition; non-elementarity paper; Lean walkthrough	Fixed notation, theorem-status conventions, and the boundary between formalized and exploratory claims.
Canonical frontier volume	`non-formalized-research-frontiers.tex`	Checked to avoid repeating the established dyadic, q-series, inverse, transform, Thue–Morse, and endpoint programs.
Frontier draft manifest	The Fourier-decay corpus and the consolidated exponents/q-series, spectra/arithmetic, integration/transforms, inverse/sampling, representations, and collected frontier volumes	Used as the recursive topic index after the 28 August 2026 reorganization; exact-title overlap was treated as a warning to narrow or redirect a proposed result.
Vendored papers	Local TeX sources for the Fabius papers and related analytic literature	Source-level comparison of normalizations, named results, and historical attributions.
Frozen archive	Early notes, first-generation primary drafts, supplements, q-calculus, walkthroughs, glossaries, standalone studies, and former frontier dossiers	Provenance and historical nonduplication only. The archive manifest explicitly identifies living or consolidated successors, so repeated text was not promoted as independent support.
Attached Russian chapter	OCR TeX plus the surviving printed PDF pages	Translation basis for the atomic-function criterion, the family h_a, the spline geometry for $a > 2$, the critical function up, dyadic values, polynomial reproduction, and the opening of the Fup hierarchy.

The nonduplication test was formula-level rather than title-level. Before designating a result [N], its defining formula, parameter range, and conclusion were compared with the canonical primary and frontier sources and with the draft manifest. Results retained as new here are therefore concentrated in the non-dyadic geometry of h_a: the exact Cantor nonanalytic locus, exact gap degrees, derivative norms, fractal zeta and tube formulas, the critical degree law, general-base cumulant limits, and the arbitrary-base Gamma–zeta periodic renormalization. This procedure cannot establish historical priority beyond the audited repository and cited literature; it does establish that the report does not intentionally duplicate an exact theorem already indexed in the repository snapshot.

B OCR reconstruction ledger

The following recurring OCR errors were normalized.

OCR pattern	Reconstruction	Reason
`ta-j`, `a-i`	ta^{-j}, a^{-j}	Geometric Fourier scales and the printed superscripts.
`l l`, `j j`	$l!$, $j!$	Taylor denominators, confirmed by neighboring remainder formulas.
`C_n^2` in exponents	$\binom{n}{2}$	Source notation for triangular exponents.
up, Cyrillic glyphs resembling "ir, separated letters	up	Cyrillic/Latin OCR mixing.
`Pj(k)`	$p_j(k)$	The jth binary digit of k.
`delta` indices shifted by one	$\delta_k = (-1)^{\mathrm{wt}_2(k-1)}$	Matches the printed recursion $\delta_{2k-1} = \delta_k$, $\delta_{2k} = -\delta_k$.
Damaged (S3.54)–(S3.58)	finite moment reduction at cylinder endpoints	The page image determines the structure but not every superscript reliably; no false literal transcription was attempted.
Damaged (S3.73)–(S3.75)	binary recursion plus error $\leq$ $\mathrm{up}(-1 + 2^{-N})$	The iterative identity and remainder are clear; the displayed digit-index series is not fully recoverable from OCR alone.
Uniqueness growth denominator	$2^{(n+1)(n+2)/2}$	Confirmed by the source's counterexample $\mathrm{up} + \alpha\,\mathrm{up}'$ and its limiting ratio.

C Notation concordance

Symbol	Meaning here	Relation to source/repository
h_a	density with Fourier product $\prod_{j\geq1} \mathrm{sinc}(ta^{-j})$	Source notation; $h_2 = \mathrm{up}$.
Φ_a	characteristic function of X_a	Source often writes $F_a(t)$; renamed to avoid conflict with bounded Fabius F.
F	bounded Fabius function on $[0, 1]$ with exterior constants	Primary repository convention.
up	even compactly supported Rvachev bump	Source up.
X_a	$\sum a^{-j} U_j$	Source random variable $\xi(a)$.
b	support radius $1/(a-1)$	Source (S3.27).
b_1	central plateau half-width $(a-2)/(a(a-1))$	Source immediately before (S3.40).
K_a	exact nonanalytic Cantor set for $a > 2$	New precise identification of the source's sing_a.
δ_k	$(-1)^{\mathrm{wt}_2(k-1)}$	Source recursive sign.
$c_{2m}(a)$	Taylor coefficient of Φ_a	Source (S3.29)–(S3.32).
$\mu_{2m}(a)$	even moment of h_a	Source (S3.36)–(S3.38).
P_a	general-base periodic Laplace correction	Generalizes the primary repository's dyadic correction.

D Pseudocode for exact gap geometry

```
input: base a>2, maximum generation N
b1 = (a-2)/(a*(a-1))
gaps = [(-b1,b1)]
for n = 0,...,N:
    output every interval in gaps with degree n
    next = []
    for (left,right) in gaps:
        next.append(((left-1)/a,(right-1)/a))
        next.append(((left+1)/a,(right+1)/a))
    gaps = sort(next)
```

For moments, compute cumulants from (13) and use the Bell recurrence

$$\mu_0 = 1, \qquad \mu_n = \sum_{j=1}^{n} \binom{n-1}{j-1} \kappa_j \mu_{n-j}. \tag{84}$$

E Reproducibility manifest

The archive contains:

- this LaTeX source and its compiled PDF;
- `experiments.py`, with detailed comments and a fixed random seed;
- vector and raster versions of all five figures;
- CSV files for geometry and derivative norms, moment validation, periodicity residuals, and Gamma–zeta Fourier-mode validation;
- `README.md`, `requirements.txt`, and `SHA256SUMS`.

Run

```
python experiments.py --output .
latexmk -pdf Atomic_Functions_Beyond_Dyadic.tex
```

from the report directory. The figures are deterministic; the Monte Carlo table is reproducible because the seed is fixed.

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